

Chapter 4 The statistical interpretation of entropy

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Review

- ◆ Gibbs described the entropy of a system as being a measure of its “**degree of mixed-up-ness**” at the atomic or molecular level. We can understand this concept of mixed-up-ness to be applied at the atomic or molecular level; that is, **the more “mixed up” the constituent particles of a system, the larger is the value of its entropy.**
- ◆ crystalline solid state vs liquid state vs gaseous state
- ◆ The development of a **quantitative relationship** between entropy and the degree of mixed-up-ness requires quantification of the term degree of mixed-up-ness, and this can be obtained from a consideration of **elementary statistical thermodynamics**. Statistical thermodynamics postulates that the equilibrium state of a system is simply the most probable of all of its possible (i.e., accessible) **microstates**.

Isolated systems, where $S=S(U,V,N)$ or $U=U(S,V,N)$
microcanonical

Review

(1) Identical Particles on *Distinguishable* Sites with Different Assigned Energies

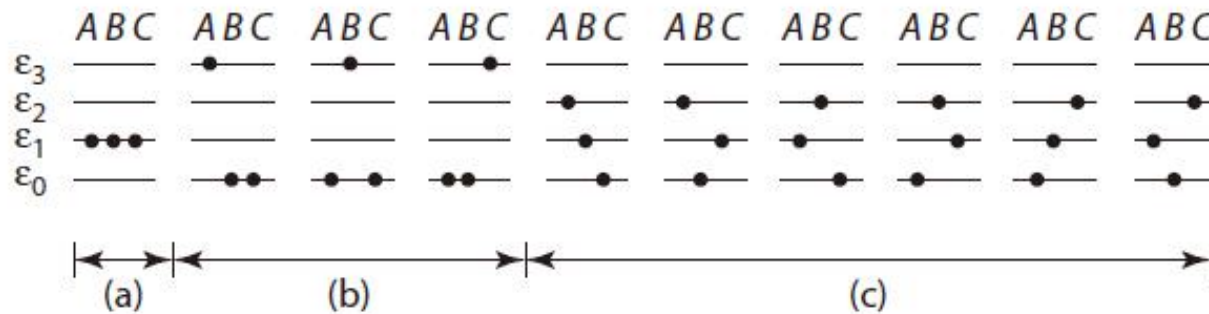


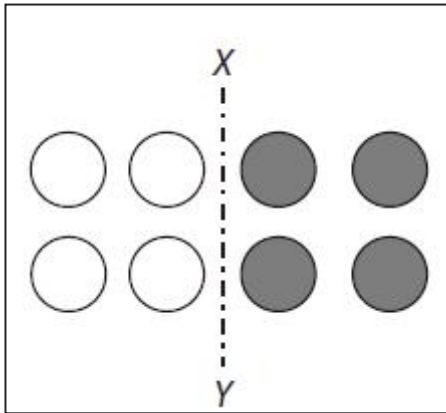
Figure 4.2 Illustration of the complexions or microstates within distributions of particles among energy levels in a system of constant energy.

- ◆ Since the equilibrium state in a closed system is the one of greatest entropy, the entropy, S , varies with Ω : $S \sim \Omega$

Hence, the entropy of an isolated system is calculated from the microstates which maximize the spread of the particles among the energy levels.

Review

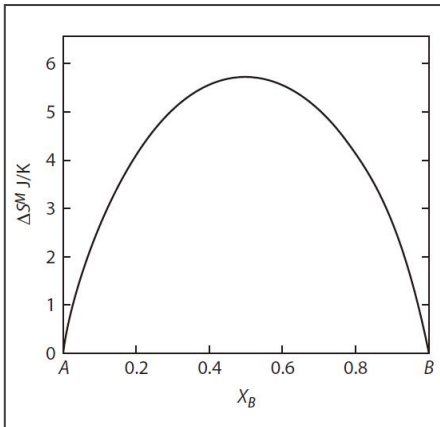
(2) Configurational Entropy of Differing Atoms in a Crystal



Atoms on Left of Partition	Atoms on Right of Partition	No. of Ways (Complexions)
4 white	4 gray	1
3 white/1 gray	3 gray/1 white	16
2 white/2 gray	2 gray/2 white	36
1 white/3 gray	1 gray/3 white	16
4 gray	4 white	1

$$\Omega = \frac{N!}{n! \times (N-n)!}$$

$A + B(\text{umixed}) \rightarrow A + B(\text{mixed})$ at constant U , V , and n



$$\Delta S'_{\text{conf}} = k_B \ln \frac{(n_a + n_b)!}{n_a! n_b!}$$

$$\Delta S_{\text{conf}} = R (X_A \ln X_A + X_B \ln X_B)$$

Review

(3) Configurational Entropy of Magnetic Spins on an Array of Atoms

- ◆ Consider a collection of magnetic spins in a paramagnet, i.e., one in which the spins are randomly placed on the atoms. The spins may take values of $\pm 1/2$. The spins will be called *up* and *down* spins for convenience, even though they do not have a direction in space.

The magnetization of a material is defined as

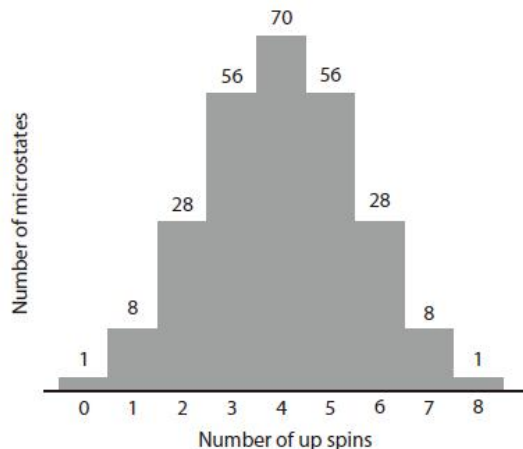


Figure 4.5 Plot of the number of microstates against the number of up spins in each microstate.

$$M = \sum_i m_i$$

The magnetization

The sum of the moments of the $\pm$ spins

4.5 The Boltzmann Distribution

- ◆ The number of arrangements within a given distribution, Ω , is calculated as follows: If n particles are distributed among the energy levels such that n_0 are in level ε_0 , n_1 in level ε_1 , n_2 in level ε_2 ..., and n_r in the highest level of occupancy ε_r , then the number of arrangements, Ω , is given by:

$$\Omega = \frac{n!}{n_0! n_1! n_2! \dots n_r!} = \frac{n!}{\prod_{i=0}^{i=r} n_i!}$$

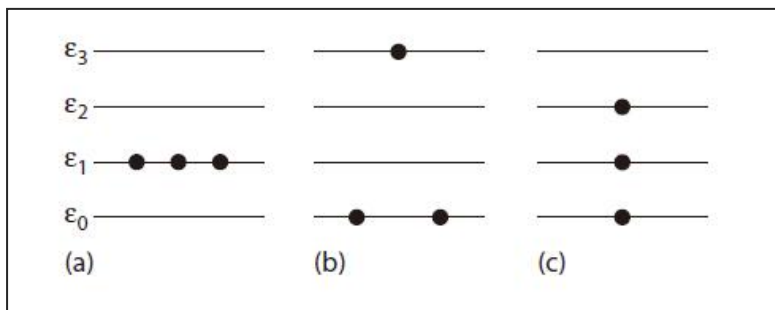


Figure 4.1 The distributions of particles among energy levels in a system of constant energy

$$\Omega(\text{distribution a}) = \frac{3!}{3!0!0!} = 1$$

$$\Omega(\text{distribution b}) = \frac{3!}{2!1!0!} = 3$$

$$\Omega(\text{distribution c}) = \frac{3!}{1!1!1!} = 6$$

4.5 The Boltzmann Distribution

- ◆ The most probable distribution is obtained by determining the set of numbers $n_0, n_1 \dots n_r$ which maximizes the value of Ω . When the values of n_i are large, Stirling's approximation can be used (that is $\ln X! = X \ln X - X$). Thus, taking the logarithms of the terms:

$$\ln \Omega = n \ln n - n - \sum_{i=0}^{i=r} (n_i \ln n_i - n_i)$$

- ◆ Since the macrostate of the system is determined by the fixed values of U , V and n , any distribution of the particles among the energy levels must conform with the conditions

$$U = \text{constant} = \sum_{i=0}^{i=r} n_i \varepsilon_i \qquad n = \text{constant} = \sum_{i=0}^{i=r} n_i$$

4.5 The Boltzmann Distribution

$$U = \text{constant} = \sum_{i=0}^{i=r} n_i \varepsilon_i$$



$$\delta U = \sum_{i=0}^{i=r} \varepsilon_i \delta n_i = 0$$

$$n = \text{constant} = \sum_{i=0}^{i=r} n_i$$



$$\delta n = \sum_{i=0}^{i=r} \delta n_i = 0$$

$$\delta \ln \Omega = - \sum_{i=0}^{i=r} \left(\delta n_i \ln n_i + \frac{n_i}{n_i} \delta n_i - \delta n_i \right) = - \sum_{i=0}^{i=r} (\delta n_i \ln n_i)$$

- ✓ If Ω has the maximum possible value then a small rearrangement of particles among the energy levels will not alter the value of Ω or of $\ln \Omega$. Thus, if the set of n_i 's is such that has its maximum value then:

$$\delta \ln \Omega = - \sum_{i=0}^{i=r} \ln n_i \delta n_i = 0$$

4.5 The Boltzmann Distribution

- ◆ The condition that Ω has its maximum value for the given macrostate is thus that the three equations can be simultaneously satisfied. The set of values of n_i in the most probable distribution is obtained by the method of undetermined **multipliers**, in the following manner.

$$\delta \ln \Omega = - \sum_{i=0}^{i=r} \ln n_i \delta n_i = 0$$

$$\delta U = \sum_{i=0}^{i=r} \varepsilon_i \delta n_i = 0$$



$$\sum_{i=0}^{i=r} \beta \varepsilon_i \delta n_i = 0$$

$$\delta n = \sum_{i=0}^{i=r} \delta n_i = 0$$



$$\sum_{i=0}^{i=r} \alpha \delta n_i = 0$$



$$\sum_{i=0}^{i=r} (\alpha + \beta \varepsilon_i + \ln n_i) \delta n_i = 0$$

4.5 The Boltzmann Distribution

$$\sum_{i=0}^{i=r} (\alpha + \beta \varepsilon_i + \ln n_i) \delta n_i = 0$$



$$\alpha + \beta \varepsilon_i + \ln n_i = 0 \quad i = 0, 1, 2, \dots, r$$

$$n_i = e^{-\alpha} e^{-\beta \varepsilon_i} \quad i = 0, 1, 2, \dots, r$$

The summation

$$\sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} = e^{-\beta \varepsilon_0} + e^{-\beta \varepsilon_1} + e^{-\beta \varepsilon_2} + \dots + e^{-\beta \varepsilon_r} \quad i = 0, 1, 2, \dots, r$$

which is determined by the magnitude of β and by the quantization of the energy, is called the partition function, P .

4.5 The Boltzmann Distribution

$$n = \sum_{i=0}^{i=r} n_i = e^{-\alpha} \sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} \quad \longrightarrow \quad e^{-\alpha} = \frac{n}{P}$$
$$n_i = e^{-\alpha} e^{-\beta \varepsilon_i} \quad \longrightarrow \quad n_i = \frac{n}{P} e^{-\beta \varepsilon_i}$$

The distribution of particles in the energy levels which maximizes (i.e., the most probable distribution) is thus one in which the occupancy of the levels decreases exponentially with increasing energy.

4.5 The Boltzmann Distribution

- ◆ The shape of this distribution is shown in Fig. 4.6. The actual shape of the exponential curve (for a given system) is determined by the value of β , and β is inversely proportional to the absolute temperature, being given by:

$$\beta = \frac{1}{k_B T}$$

in which k_B is Boltzmann's constant, an expression of the gas constant per atom or molecule, i.e.,

$$\begin{aligned} k_B &= \frac{R}{N_0} = \frac{8.314}{6.0232 \times 10^{23}} \\ &= 1.38054 \times 10^{-23} \text{ J/K} \end{aligned}$$

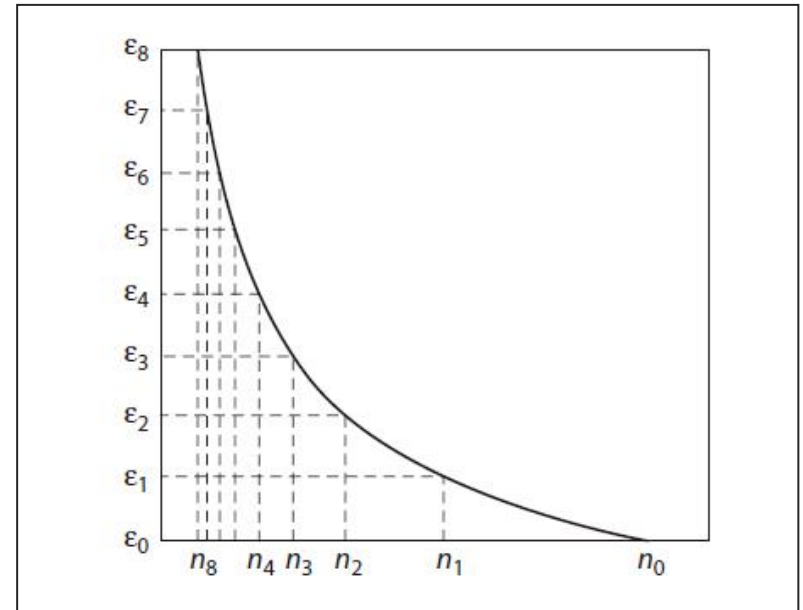


Figure 4.6 Schematic representation of the most probable distribution of particles among the quantized energy levels.

4.6 The influence of temperature

- ◆ The nature of the exponential distribution of particles is determined by the temperature of the system. However, as the macrostate of the system is fixed by fixing the values of U , V , and n , then T , as a dependent variable, is fixed.
- ◆ $\beta=1/k_B T$ shows that T increases with decreasing β . An increase in temperature causes the upper energy levels to become relatively more populated, and this corresponds to an increase in the average energy of the particles, i.e., to an increase in the value of U/n , which, for fixed values of V and n , corresponds to an increase in U .

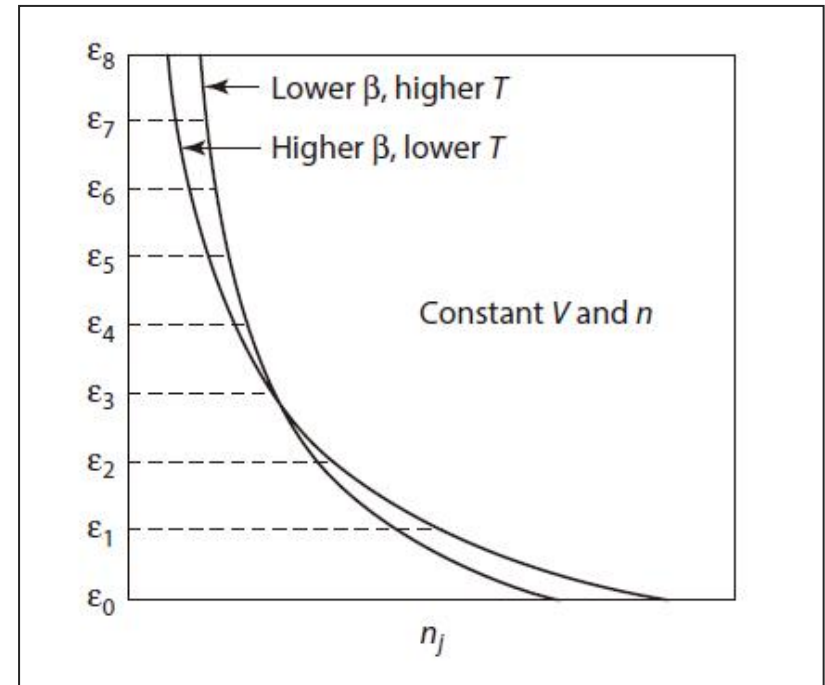


Figure 4.7 The influence of temperature on the most probable distribution of particles among energy levels in a closed system of constant volume.

4.6 The influence of temperature

- ◆ As has been stated, when the number of particles in the system is very large, the number of arrangements within the most probable distribution, $\Omega_{\max}$, is the only term which makes a significant contribution to the total number of arrangements, Ω_{total} , which the system may have; that is, $\Omega_{\max}$ is significantly larger than the sum of all of the other arrangements. Thus, when the number of particles is large, Ω_{total} can be equated with $\Omega_{\max}$.

$$\ln \Omega_{\text{total}} = \ln \Omega_{\max} = n \ln n - n - \sum_{i=0}^{i=r} (n_i \ln n_i - n_i) = n \ln n - \sum_{i=0}^{i=r} n_i \ln n_i$$

$$n_i = \frac{n}{P} e^{-\beta \varepsilon_i} = \frac{n}{P} e^{-\varepsilon_i / k_B T}$$

4.6 The influence of temperature

$$\begin{aligned}\ln \Omega_{\text{total}} &= n \ln n - \sum_{i=0}^{i=r} n_i \ln n_i \\&= n \ln n - \sum_{i=0}^{i=r} \frac{n}{P} e^{-\varepsilon_i/k_B T} \ln \frac{n}{P} e^{-\varepsilon_i/k_B T} \\&= n \ln n - \frac{n}{P} \sum_{i=0}^{i=r} \left[e^{-\varepsilon_i/k_B T} \left(\ln n - \ln P - \frac{\varepsilon_i}{k_B T} \right) \right] \\&= n \ln n - \frac{n}{P} (\ln n - \ln P) \sum_{i=0}^{i=r} e^{-\varepsilon_i/k_B T} + \frac{n}{P k_B T} \sum_{i=0}^{i=r} \varepsilon_i e^{-\varepsilon_i/k_B T} \\U = \text{constant} &= \sum_{i=0}^{i=r} \frac{n \varepsilon_i e^{-\varepsilon_i/k_B T}}{P} = \frac{n}{P} \sum_{i=0}^{i=r} \varepsilon_i e^{-\varepsilon_i/k_B T}\end{aligned}$$

4.6 The influence of temperature

$$\begin{aligned}\ln \Omega_{\text{total}} &= n \ln n - \frac{n}{P} (\ln n - \ln P) \sum_{i=0}^{i=r} e^{-\varepsilon_i/k_B T} + \frac{n}{P k_B T} \sum_{i=0}^{i=r} \varepsilon_i e^{-\varepsilon_i/k_B T} \\ &= n \ln n - \frac{n}{P} (\ln n - \ln P) P + \frac{n}{P k_B T} \frac{UP}{n} \\ &= n \ln P + \frac{U}{k_B T}\end{aligned}$$

$$k_B T \ln \Omega_{\text{total}} = n k_B T \ln P + U$$

Helmholtz free energy (Chapter 5): $A = U - TS$

$$A = -n k_B T \ln P$$

$$S = k_B \ln \Omega_{\text{total}}$$

4.7 Thermal equilibrium and the Boltzmann equation

- ◆ Consider now a system of particles in thermal equilibrium with a heat bath. Let the state of the combined system (**particles + heat bath**) be fixed by fixing the values of U , V , and n , where

$$U = U_{\text{particle system}} + U_{\text{heat bath}}$$

$$V = V_{\text{particle system}} + V_{\text{heat bath}}$$

n = number of particles in the system + the heat bath of fixed size

4.7 Thermal equilibrium and the Boltzmann equation

- ◆ As the particles system and the heat bath are in thermal equilibrium, small exchanges of energy can occur between them, and for such a small exchange at constant U , V , and n , we have:

$$\ln \Omega_{total} = n \ln P + \frac{U}{k_B T} \quad \Rightarrow \quad \delta \ln \Omega_{total} = \frac{dU}{k_B T}$$

P is dependent only on the values of ϵ_i and T

- ◆ As this exchange of energy is carried out at constant total volume, then

$$\delta \ln \Omega_{total} = \frac{dU}{k_B T} = \frac{\delta q_V}{k_B T} = \frac{dS}{k_B}$$

$$dS = k_B \delta \ln \Omega$$

4.7 Thermal equilibrium and the Boltzmann equation

$$dS = k_B \delta \ln \Omega$$

- ◆ As both S and Ω are state functions, the above expression can be written as a differential equation, integration of which gives:

$$\Delta S = k_B \ln \Omega$$

- ◆ This is the Boltzmann's equation, is the required quantitative relationship between the entropy of a system and its “degree of mixed-up-ness,” in which the latter, given by Ω , is the number of ways in which the energy of the system can be distributed among the particles.
- ◆ The most probable state of the system is that in which Ω has a maximum value, consistent with the fixed values of U , V , and n , and hence the equilibrium state of the system is that in which S is a maximum, consistent with the fixed values of U , V , and n . The Boltzmann equation thus provides a physical quality to entropy.

4.8 Heat flow and the production of entropy

- ◆ Classical Thermodynamics shows that the transfer of heat from a body at some temperature to a body at a lower temperature is an irreversible process which is accompanied by the production of entropy, and that the reverse process, i.e., the flow of heat up a temperature gradient, is an impossible process.
- ◆ An examination of microstates shows that a microstate in which variations in temperature occur within a system is less probable than a microstate in which the temperature of the system is uniformly constant.

4.8 Heat flow and the production of entropy

- ◆ Consider two closed systems, A and B. Let the energy of A be U_A and the number of complexions of A be Ω_A . Similarly let the energy of B be U_B and its number of complexions be Ω_B .
- ◆ When thermal contact is made between A and B, the product $\Omega_A\Omega_B$ will, generally, not have its maximum possible value, and heat will be transferred either from A to B or from B to A. Heat flows from A to B if, thereby, the increase in Ω_B , caused by the increase in U_B , is greater than the decrease in Ω_A , caused by the decrease in U_A .
- ◆ Heat continues to flow from A to B as long as the product $\Omega_A\Omega_B$ continues to increase, and the flow of heat ceases when $\Omega_A\Omega_B$ reaches its maximum value, i.e., when the increase in Ω_B caused by the transfer of an increment of heat from A is exactly compensated by the decrease in Ω_A .
- ◆ The condition for A to be in thermal equilibrium with B is thus that the transfer of a quantity of heat from one body to the other does not cause a change in the value of $\Omega_A\Omega_B$. That is,
$$\delta \ln \Omega_A \Omega_B = 0$$

4.8 Heat flow and the production of entropy

- ◆ Consider a rearrangement of the particles in the quantized energy levels in B which causes U_B to increase by a certain amount, and consider a simultaneous rearrangement of the particles in the energy levels of A which causes U_A to decrease by the same amount; i.e., $(U_A + U_B)$ remains constant.

$$\delta \ln \Omega_{total} = \frac{dU}{k_B T}$$
$$\delta \ln \Omega_A = \frac{dU_A}{k_B T_A} = \frac{\delta q_A}{k_B T_A}$$
$$\delta \ln \Omega_B = \frac{dU_B}{k_B T_B} = \frac{\delta q_B}{k_B T_B}$$

4.8 Heat flow and the production of entropy

- ◆ When a quantity of heat is transferred from A to B at total constant energy, then

$$\delta q_A = -\delta q_B$$

$$\delta \ln \Omega_A \Omega_B = \delta \ln \Omega_A + \delta \ln \Omega_B = \frac{\delta q_A}{k_B T_A} + \frac{\delta q_B}{k_B T_B} = \frac{\delta q_A}{k_B} \left(\frac{1}{T_A} - \frac{1}{T_B} \right)$$

- ◆ The condition that $\delta \ln \Omega_A \Omega_B$ be zero is that $T_A = T_B$. The reversible transfer of heat from one body to another thus only occurs when the temperatures of the bodies are equal, as only in such a case $\Omega_A \Omega_B$, and hence the total entropy of the combined system ($S_A + S_B$), remain constant. An irreversible transfer of heat increases the value of the product $\Omega_A \Omega_B$, and hence entropy is created.

4.8 Heat flow and the production of entropy

- ◆ From the point of view of microstates, an irreversible process is one which takes the system from **a less probable state** to **the most probable state**, and from the point of view of macrostates, an irreversible process takes the system from a **nonequilibrium state** to the **equilibrium state**.
- ◆ What is considered in Classical Thermodynamics to be an impossible process is shown by a consideration of microstates to only be an improbable process.
- ◆ As the size of the system increases, the probability of the “up-hill” flow of thermal energy approaches zero. For very small systems (clusters of atoms), these improbable processes must be considered.

4.8 Heat flow and the production of entropy

- ◆ The total entropy of a simple system consists of its thermal entropy, S_{th} , which arises from the number of ways in which the energy of the system can be shared among the particles, and its configurational entropy, S_{conf} , which arises from the number of distinguishable ways in which the particles can fill the space available to them.

$$S_{total} = S_{th} + S_{conf} = k_B \ln \Omega_{th} + k_B \ln \Omega_{conf} = k_B \ln \Omega_{th} \Omega_{conf}$$

- ◆ If two systems are placed together in thermal contact, the change in entropy is given as:

$$\Delta S_{total} = k_B \ln \frac{\Omega_{th(2)} \Omega_{conf(2)}}{\Omega_{th(1)} \Omega_{conf(1)}}$$

4.8 Heat flow and the production of entropy

$$\Delta S_{total} = k_B \ln \frac{\Omega_{th(2)} \Omega_{conf(2)}}{\Omega_{th(1)} \Omega_{conf(1)}}$$

If, however, the two systems were closed systems, there can be no change in the configurational entropy: $\Omega_{conf(1)} = \Omega_{conf(2)}$. This is also true if the two systems were open systems which were chemically identical. Thus, in the case of thermal energy flow down a temperature gradient between two such systems, since only Ω_{th} changes, the increase in the entropy arising from the thermal energy transfer which takes the system from state 1 to state 2 is:

$$\Delta S_{total} = k_B \ln \frac{\Omega_{th(2)} \Omega_{conf(2)}}{\Omega_{th(1)} \Omega_{conf(1)}} = k_B \ln \frac{\Omega_{th(2)}}{\Omega_{th(1)}} = \Delta S_{th}$$

4.8 Heat flow and the production of entropy

- ◆ In the mixing of particles of A with particles of B , ΔS_{total} only equals ΔS_{conf} if the mixing process does not cause a redistribution of the particles among the energy levels—that is, if $\Omega_{\text{th}(1)} = \Omega_{\text{th}(2)}$. This condition corresponds to the *ideal mixing* of the particles and requires that the quantization of energy be the same in crystals A and B .
- ◆ Ideal mixing is the exception rather than the rule, and, generally, when two or more components are mixed at constant U , V , and n , $\Omega_{\text{th}(2)}$ does not have the same value as $\Omega_{\text{th}(1)}$; thus, the completely random mixing of particles does not occur. In such cases, either the clustering of like particles (indicating difficulty in mixing) or ordering (indicating a tendency toward compound formation) occurs.
- ◆ In all cases, however, the equilibrium state of the system is that which, at constant U , V , and n , maximizes the product $\Omega_{\text{th}}\Omega_{\text{conf}}$. If, in addition to particles, the system is composed of magnetic spins, electric dipoles, and so on, their entropy must be included in the consideration of the total entropy of the system.

4.9 Summary

1. A single macrostate of a system, which is determined when the independent variables of the system are fixed, contains a very large number of microstates, each of which is characterized by the manner in which

- The thermal energy of the system is distributed among the particles
- The particles are distributed in the configurations available to them

2. Although the occurrence of a system in any one of its microstates is equally probable (for the case of the closed system), greatly differing numbers of microstates occur in differing distributions. The distribution which contains the largest number of microstates is the most probable distribution.

4.9 Summary

3. In real systems (with a very large number of particles) the number of microstates in the most probable distribution is significantly larger than the sum of all of the other microstates occurring in all of the other distributions. This most probable distribution is the equilibrium thermodynamic state of the system.

4. The relationship between the number of microstates available to the system, Ω , and the entropy of the system is given by the Boltzmann equation:

$$S = k_B \ln \Omega$$

in which k_B is Boltzmann's constant

4.9 Summary

5. If a situation arises which allows an increase in the number of microstates available to the system (at constant energy), then spontaneous redistribution of the energy among the particles (or particles over the available configurations) occurs until the newly available most probable distribution occurs. The Boltzmann equation shows that an increase in the number of microstates made available to the system causes an increase in the entropy of the system.

6. The total entropy of a system is the sum of each of the aspects of entropy:

- Thermal entropy, S_{th}
- Configurational entropy, S_{conf}
- Any other aspect of entropy, such as spin entropy, electric dipole entropy, and so on

4.9 Summary

7. The thermal entropy arises from the number of ways in which the thermal energy available to the system can be shared among the constituent particles, Ω_{th} .
8. The configurational entropy arises from the number of ways in which the particles can be distributed over the configurations available to them, Ω_{conf} .
9. The spin entropy arises from the number of ways in which the spins can be distributed over the sites available to them, Ω_{spin} .
10. Since any of the thermal distributions can be combined with any of the configurational distributions, the total number of microstates available to the system is the product $\Omega_{\text{th}} \Omega_{\text{conf}}$ ($\Omega_{\text{th}} \Omega_{\text{conf}} \Omega_{\text{spin}}$ if spins are present), and hence, from the logarithmic form of the Boltzmann equation, the total entropy of the system is the sum of the individual entropies of the system.

4.10 Concepts and Terms Introduced

Aspects of entropy

Boltzmann distribution

Boltzmann equation for entropy

Bond energy

Classical thermodynamics

Configurational entropy

Entropy as “mixed-up-ness”

Exchange field

External field

Thermal entropy

Macrostate

Microcanonical ensemble

Microstate

Most probable state

Paramagnet

Partition function

Phenomenological

Spin entropy

Spin system

Statistical thermodynamics

4.11 Qualitative Example Problems

Qualitative Problem 1

One mole of a FeX alloy of 50 at% X has its Fe and X atoms randomly arranged on a body centered cubic lattice (BBC). The spins on the Fe atoms are also randomly arranged. The X atoms do not have spins.

- Calculate the configurational entropy of the alloy.
- Calculate the spin entropy of the alloy.
- Is the sum of the entropies for (a) and (b) the total entropy of the alloy? Explain.

4.11 Qualitative Example Problems

Solution to Qualitative Problem 1

a

$$\Delta S_{conf} = k_B \ln \frac{(n_{Fe} + n_X)!}{n_{Fe}! n_X!}$$

$$\Delta S_{conf} = k_B \left[(n_{Fe} + n_X) \ln (n_{Fe} + n_X) - (n_{Fe} + n_X) - (n_{Fe} \ln n_{Fe} - n_{Fe} + n_X \ln n_X - n_X) \right]$$

$$\Delta S_{conf} = k_B (n_{Fe} + n_X) \left[\ln (n_{Fe} + n_X) - \frac{n_{Fe}}{n_{Fe} + n_X} \ln n_{Fe} - \frac{n_X}{n_{Fe} + n_X} \ln n_X \right]$$

$$n_{Fe} + n_X = 1 \quad X_{Fe} = 0.5 \quad X_X = 0.5$$

$$\Delta S_{conf} = -R [X_{Fe} \ln X_{Fe} + X_X \ln X_X]$$

c No, there is also thermal entropy, which must be included.

b

$$\Delta S_{spin} = k_B \ln 2^{n_{Fe}} = n_{Fe} k_B \ln 2 = \frac{R}{2} \ln 2$$

4.11 Qualitative Example Problems

Qualitative Problem 2

We saw in Section 4.6 that $A = -nk_B T \ln P$

In the next chapter, we will show that $\partial A / \partial T = -S$

Take the derivative of A with respect to T and obtain expressions for S and U in terms of the partition function P .

Solution
$$\frac{\partial A}{\partial T} = -nk_B \ln P - nk_B T \frac{\partial \ln P}{\partial T} = -S$$

$$nk_B \ln P + nk_B T \frac{\partial \ln P}{\partial T} = S$$

$$A = U - TS$$

$$A = -nk_B T \ln P$$

$$-nk_B T \ln P = U - T \left(nk_B \ln P + nk_B T \frac{\partial \ln P}{\partial T} \right)$$

$$U = nk_B T^2 \frac{\partial \ln P}{\partial T}$$

4.12 Qualitative Example Problems

Quantitative Problem 1

Spectroscopic observation of molecular N_2 in an electrical discharge shows that the relative numbers of molecules in excited vibrational states with energies given by

$$\varepsilon_i = \left(i + \frac{1}{2} \right) h\nu \quad (4.24)$$

are

i	0	1	2	3
$\frac{n_i}{n}$	1.00	0.250	0.062	0.016

Show that the gas is in thermodynamic equilibrium with respect to the distribution of vibrational energy, and calculate the temperature of the gas. In Equation 4.24, i is an integer which has values in the range zero to infinity, h is Planck's constant of action ($= 6.6252 \times 10^{-34} \text{ J}\cdot\text{s}$), and the vibration frequency, ν , is $7.00 \times 10^{13} \text{ s}^{-1}$.

Homework

4.12 Qualitative Example Problems

Quantitative Problem 2

The isotopic composition of lead in atomic percent is as follows:

Atomic Weight	Atomic Percent
204	1.5
206	23.6
207	22.6
208	52.3

Calculate the molar configurational entropy of lead.

Homework

Homework

4.1. A rigid container is divided into two compartments of equal volume by a partition. One compartment contains 1 mole of ideal gas A at 1 atm, and the other contains 1 mole of ideal gas B at 1 atm. Calculate the increase in entropy which occurs when the partition between the two compartments is removed. If the first compartment had contained 2 moles of ideal gas A , what would have been the increase in entropy when the partition was removed? Calculate the corresponding increases in entropy in each of the preceding two situations if both compartments had contained ideal gas A .

4.2. Show that, when n atoms of A and n atoms of B form a randomly mixed solution, the fraction of the total number of distinguishable complexions which occur in the most probable distribution decreases with the increasing value of n .

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4.3. Assuming that a silver– gold alloy is a random mixture of gold and silver atoms, calculate the increase in entropy when 10 g of gold are mixed with 20 g of silver to form a homogeneous alloy. The gram atomic weights of Au and Ag are, respectively, 198 and 107.9.

4.4. On the assumption that copper– nickel alloys are random mixtures of copper and nickel atoms, calculate the mass of copper which, when mixed with 100 g of nickel, causes an increase in entropy of 15 J/K. The gram atomic weights of Cu and Ni are, respectively, 63.55 and 58.69.

Thermodynamics of Materials & Phase transformations

The End

